

Pericardial Diseases: Approach to a Case of Chest Pain

Dr. Muhammad Reihan, Internal Medicine | Source: Reihan_pericardial_disease.pdf (33 slides)

Legend: ★ = high-yield · [added] = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

🔗 Bridge from earlier lectures:

- **CAD:** acute pericarditis is the key **STEMI mimic**. Post-MI pericarditis and **Dressler syndrome** were MI complications in that lecture's timeline.
- **Pulmonary vascular disease:** massive PE was another cause of **↑JVP + hypotension + clear lungs**.
- **Thyroid 2: hypothyroidism** causes **pericardial effusion**.
- **Pleural diseases:** pleuritic pain and a **friction rub** appeared there too. The pericardial rub persists when the patient holds their breath; the pleural rub stops.

1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	★ Clinical features of acute pericarditis , cardiac tamponade (Beck's triad) and constrictive pericarditis	Vignette → diagnosis
Knowledge	★ Management of all three	NSAID + colchicine / pericardiocentesis / pericardiectomy
Skill	★ Recognise tamponade and constriction signs on exam	Pulsus paradoxus, Kussmaul, knock, Ewart
Skill	Management plan for pericarditis, effusion/tamponade, constriction	Next best step

Lecture flow: Opening case → **Acute pericarditis** (definition, etiology, features, ECG evolution vs STEMI, labs, treatment) → **Effusion & tamponade** (definition, etiology, rate of accumulation, Beck/Ewart, ECG/CXR/echo, management, pericardiocentesis) → **Constrictive pericarditis** (definition, etiology, Kussmaul, knock, dip-and-plateau, pericardiectomy) → Summary.

2. Main Content (Lecture Order)

2.0 Opening Case (Slide 4)

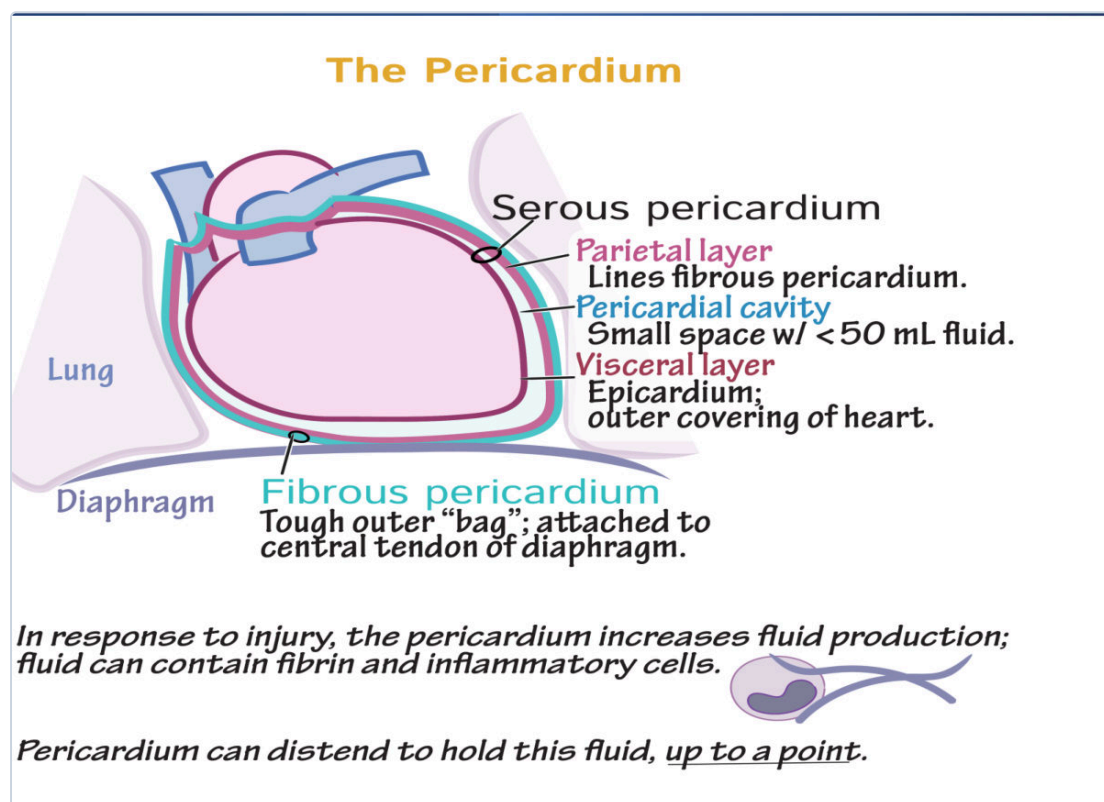
- **Patient:** 23-year-old man with **1 week** of **sharp substernal chest pain**, **worse on inspiration** and **relieved by leaning forward**.
- **History:** nausea and **myalgia**. His father has CAD.
- **Exam:** T 37.3 °C, pulse 110, BP 130/84. **High-pitched rubbing sound between S1 and S2 at the left sternal border**.
- **ECG:** **PR depression** and **diffuse ST elevation**.

→ *Most likely cause?* (a) Dressler syndrome · (b) acute MI · (c) TB · (d) SLE · (e) **acute viral infection**.
(Answered in §7, Q1.)

2.1 Anatomy Refresher (Slide 5)

Layer	Detail
Fibrous pericardium	Tough outer "bag" attached to the central tendon of the diaphragm
Serous pericardium: parietal layer	Lines the fibrous pericardium
Pericardial cavity	Small space with < 50 mL fluid
Serous pericardium: visceral layer (epicardium)	Outer covering of the heart

- In response to injury, the pericardium **increases fluid production**. The fluid can contain **fibrin and inflammatory cells**.
- The pericardium **can distend to hold fluid, up to a point**.



Slide 5: layers of the pericardium

2.2 Acute Pericarditis: Definition & Etiology (Slides 5-6)

Acute pericarditis = inflammation of the pericardium. Its exact incidence is unknown because **subclinical courses are common**.

Etiology (slide 6):

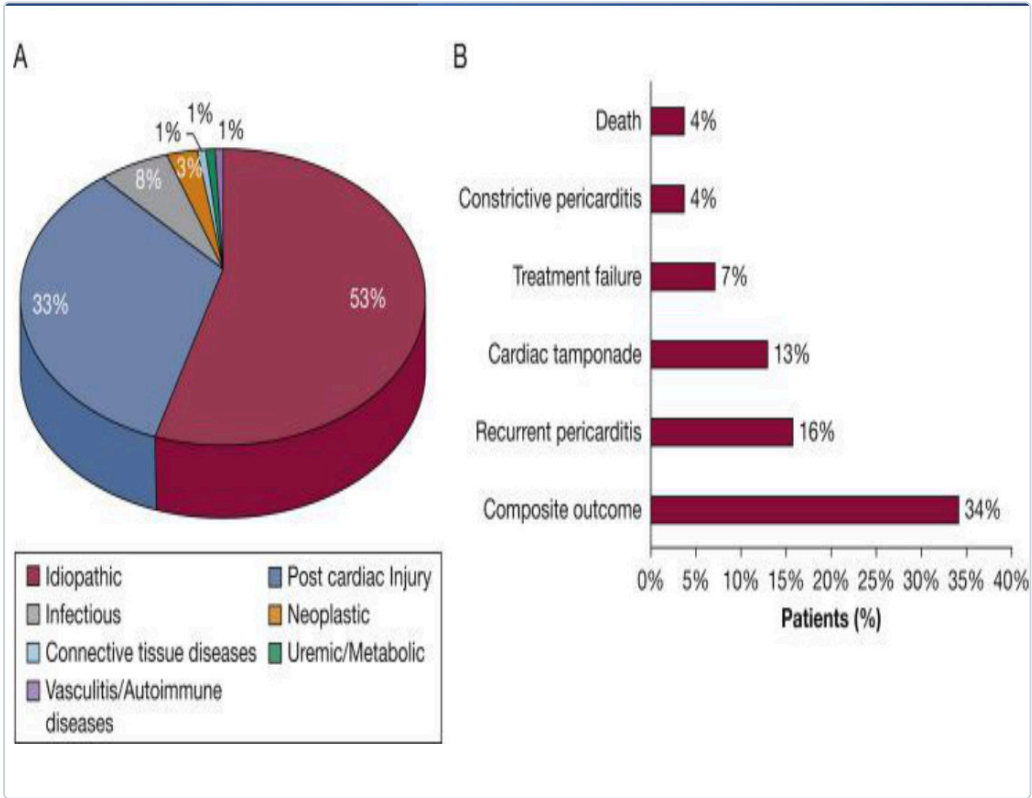
- **Idiopathic**
- ★ **Infectious (most commonly viral)**
- **Acute MI**
- **Uraemia**
- **Autoimmune connective tissue disease**

- **Neoplasms**
- **After cardiac surgery**
- **Radiation**
- **Trauma**

Slide 6 figure (cohort data, *Chest* 2020):

A. Etiology	%	B. Outcomes	%
Idiopathic	53%	Composite outcome	34%
Post-cardiac injury	33%	Recurrent pericarditis	16%
Infectious	8%	Cardiac tamponade	13%
Neoplastic	3%	Treatment failure	7%
CTD / uraemic-metabolic / vasculitis	~1% each	Constrictive pericarditis	4%
		Death	4%

The legend colours were read from the slide image. Post-cardiac injury at 33% reflects that study's cohort (hospitalised, many post-surgical). In the general population, idiopathic/viral causes dominate (~80-90%) [added]. Recurrence (16%) is the most common single complication. This is why colchicine matters.



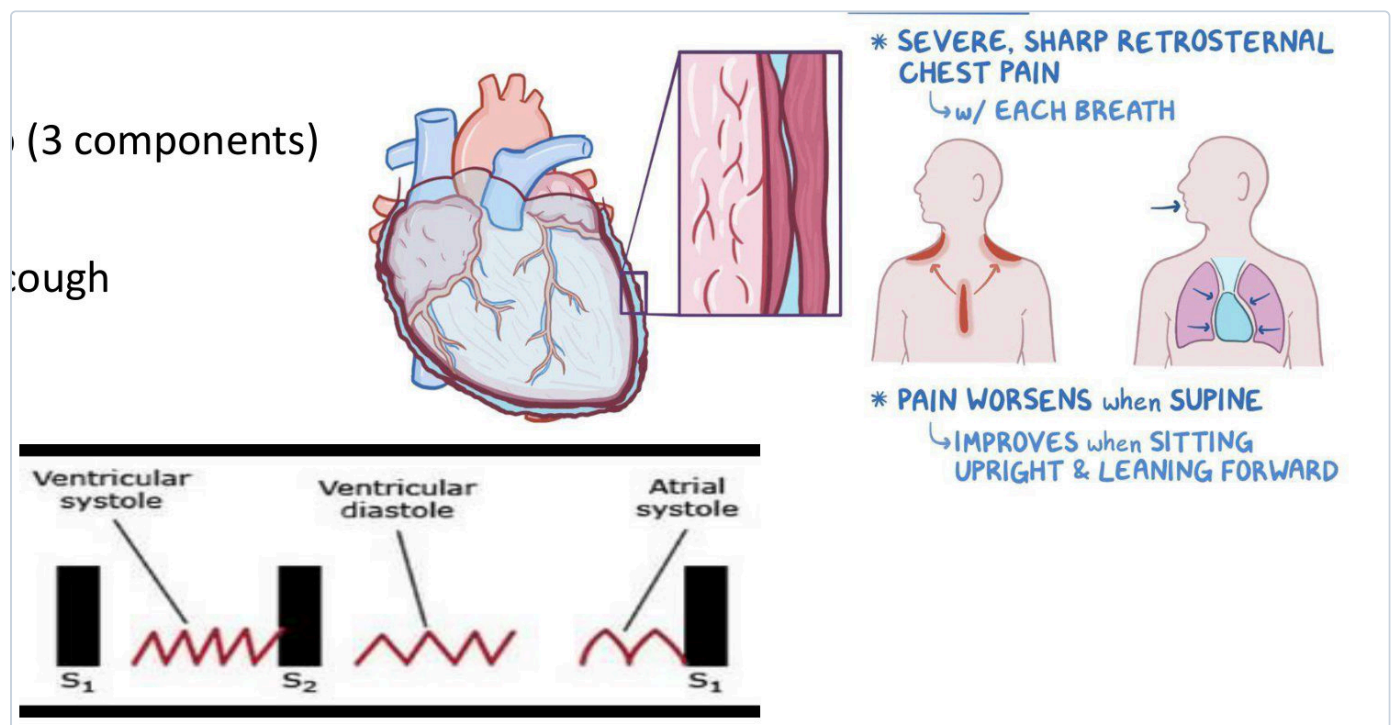
Slide 6: etiology and outcomes of acute pericarditis

Mnemonic for causes: "CARDIAC RIND" [added]

- **C**ollagen vascular
- **A**ortic dissection
- **R**adiation
- **D**rugs (hydralazine, procainamide, isoniazid)
- **I**nfection
- **A**cute renal failure (uraemia)
- **C**ardiac (MI, Dressler)
- **R**heumatic fever
- **I**njury
- **N**eoplasm
- **D**ressler

2.3 Clinical Features (Slide 7) ★

- ★ **Positional chest pain**: - Severe, sharp, retrosternal, **worse with each breath** and **when supine**. - **Better sitting upright and leaning forward**.
- ★ **Pericardial friction rub**: **3 components**: - **Ventricular systole** - **Ventricular diastole** - **Atrial systole**
- Low-grade fever
- Dyspnoea, tachypnoea, cough
- **Hiccups** ([added] phrenic nerve runs over the pericardium)
- Pericardial effusion



Slide 7: symptoms and the 3-component friction rub

[added] The rub is high-pitched and scratchy, best heard at the left lower sternal border, leaning forward, at end-expiration. It is often transient, so listen repeatedly. Pain radiating to the trapezius ridge is highly specific (phrenic nerve).

2.4 Diagnostics (Slides 8-12) ★

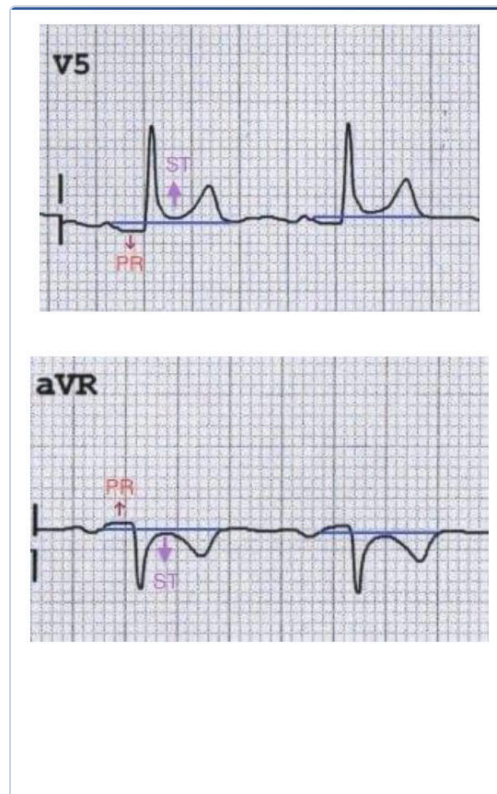
ECG:

- Changes **evolve over days to weeks**.
- **Widespread ST elevation + PR depression**.
- **Reciprocal ST depression + PR elevation in aVR and V1**.

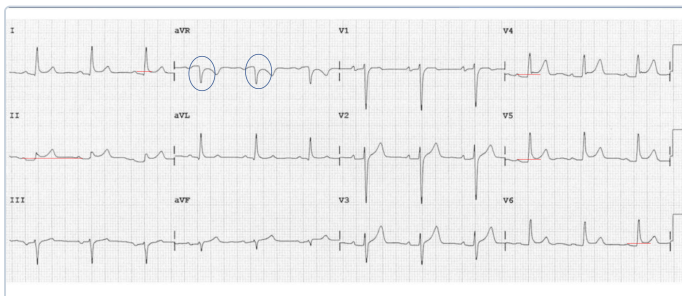
Labs:

- **Leukocytosis**
- **↑ Troponin I (minimal)**
- **↑ ESR, ↑ CRP**
- **↑ CK**

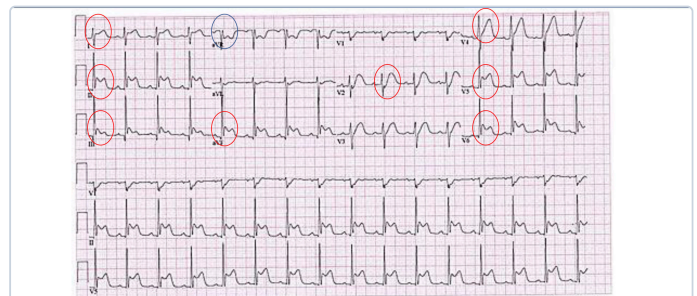
Echocardiogram to look for effusion.



Slide 8: V5 (ST↑, PR↓) vs aVR (reciprocal)



Slide 9: 12-lead ECG of acute pericarditis



Slide 11: diffuse ST elevation with PR depression (circled)

Pericarditis vs STEMI (slide 12) ★

	ST shape	Which leads?	ST-T evolution	PR segment
Pericarditis	Concave ("smiley" coving)	Diffuse	Days-weeks	Depressed
Acute STEMI	Convex (dome/tombstone)	Localised (coronary territory) + reciprocal ST↓	Hours	Normal

	ST Coving	Which Leads?	ST-T Evolution	PR Segment
Pericarditis		Diffuse	Days-weeks	Depressed
Acute STEMI		Localized	Hours	Normal

Slide 12: pericarditis vs STEMI on ECG

[added] **Four ECG stages of pericarditis:**

1. Diffuse ST↑ + PR↓
2. ST normalises, T flattens
3. **T-wave inversion** (only *after* ST returns to baseline, unlike MI)
4. Normalisation

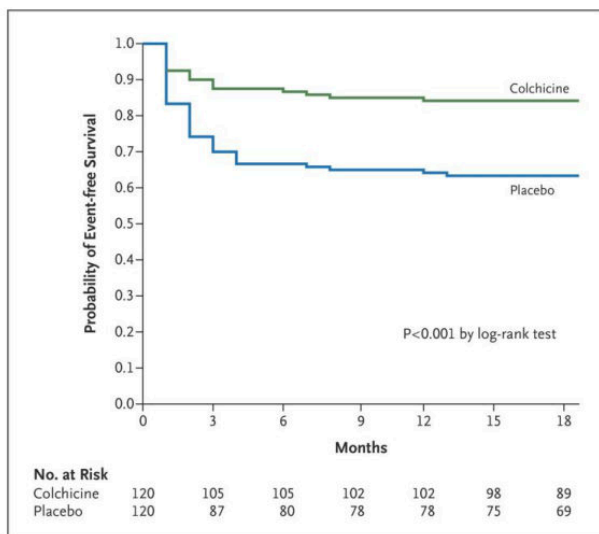
Spodick sign = down-sloping TP segment. In STEMI, **Q waves** develop and T waves invert **while ST is still elevated**.

2.5 Management (Slide 13) ★

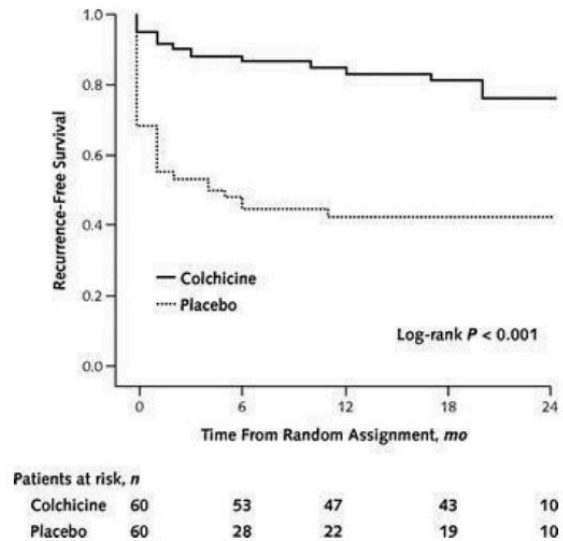
Line	Therapy
First line	★ High-dose NSAID (or aspirin) + colchicine
Second line	Glucocorticoids
Always	Gastro-protective therapy (PPI)

- **ICAP trial** (*NEJM* 2013): colchicine added to anti-inflammatory therapy **reduced recurrence** of acute pericarditis.
- **CORP trial** (*Ann Intern Med* 2011): colchicine **reduced recurrence in recurrent pericarditis**.

ICAP study



CORP study



(Colchicine for Recurrent Pericarditis) Investigators. Colchicine for recurrent pericarditis (CORP): a randomized trial. Ann Intern Med. 2011 Oct 4;155(7)

Slide 13: ICAP and CORP Kaplan-Meier curves (colchicine vs placebo)

[added]:

- Typical regimen: ibuprofen 600–800 mg TDS or aspirin 750–1000 mg TDS for 1–2 weeks, then taper. Colchicine 0.5 mg once or twice daily for **3 months**.
- **Aspirin, not NSAIDs, after MI** (NSAIDs impair scar healing).
- **Steroids increase recurrence**, so reserve them for contraindication or failure, CTD, or pregnancy.
- **Admit** if: fever > 38 °C, subacute onset, large effusion/tamponade, failure of NSAID, immunosuppression, trauma, anticoagulation, or myopericarditis (high troponin).

🔄 **Recap: Acute pericarditis.** Inflamed pericardium, mostly idiopathic or viral. Sharp pleuritic pain, better leaning forward; 3-component rub. ECG: diffuse concave ST↑ + PR↓, reciprocal in aVR/V1, evolving over days. Troponin only minimally ↑; ESR/CRP ↑. Treat with high-dose NSAID + colchicine + PPI; steroids second line. Colchicine halves recurrence (ICAP, CORP).

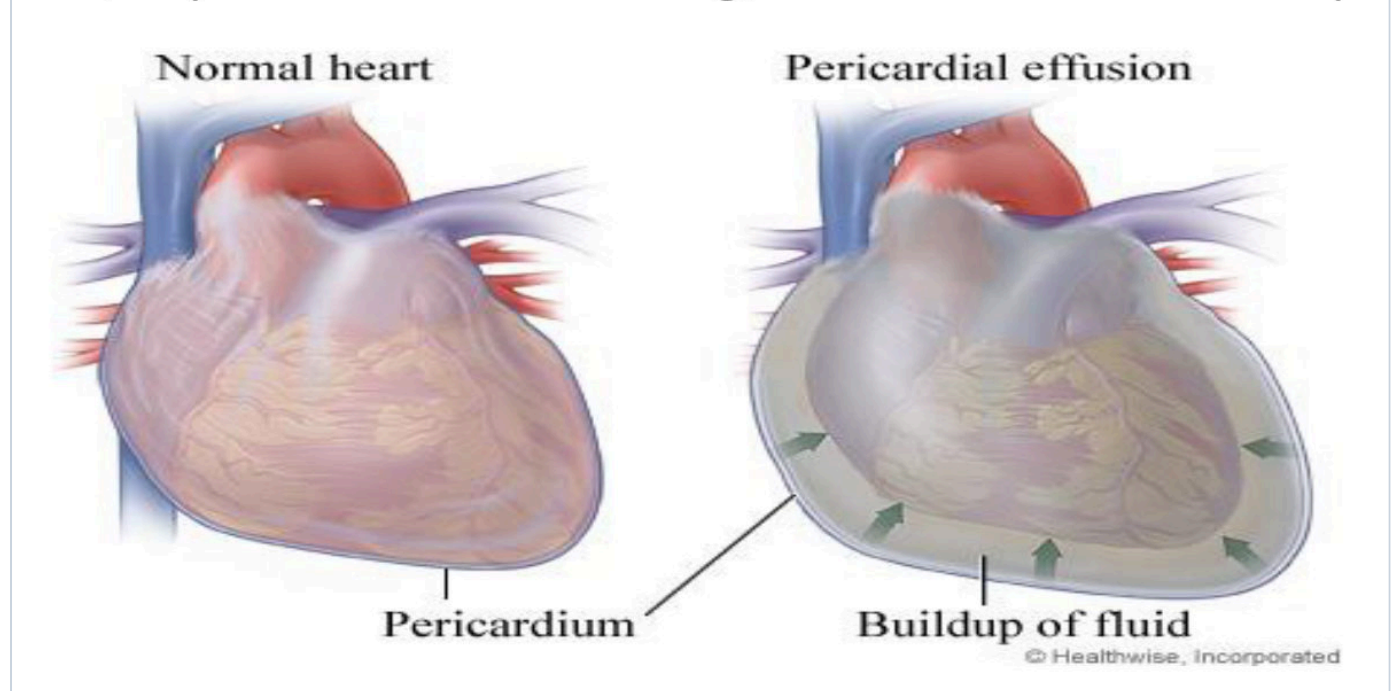
🔗 **Link:** Inflamed pericardium produces fluid. Whether that fluid is harmless or lethal depends on how fast it accumulates.

👉 **Up next: Pericardial effusion and cardiac tamponade.**

2.6 Effusion & Tamponade: Definitions (Slide 14)

- **Pericardial effusion:** abnormal collection of fluid in the pericardial space.
- ★ **Cardiac tamponade:** fluid accumulation **raises intra-pericardial pressure** → **cardiac compression** → **impaired ventricular filling** → **↓ cardiac output**.

sion, impaired ventricular filling, and reduced cardiac output



Slide 14: normal heart vs pericardial effusion

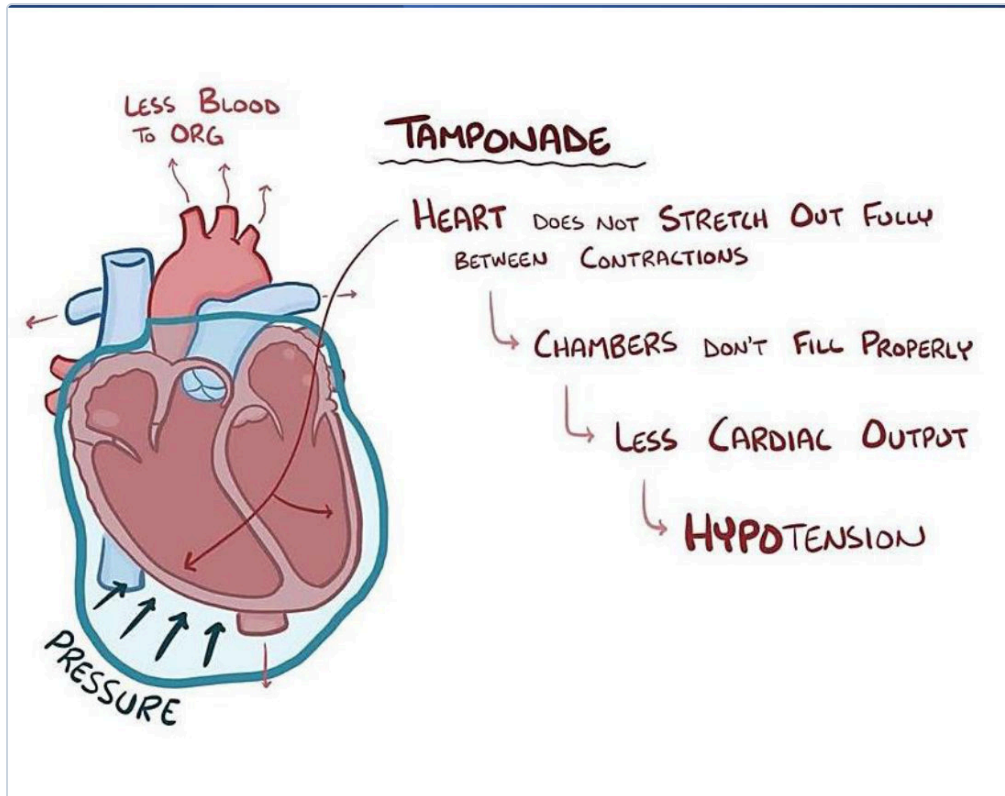
2.7 Etiology (Slide 15)

Rapid (haemopericardium, often tamponade)	Slower
Cardiac wall rupture (post-MI free wall)	Idiopathic
Chest trauma	Acute pericarditis (viral, TB , bacterial)
Aortic dissection (type A)	Malignancy
	Uraemia
	Autoimmune disorders
	★ Hypothyroidism

[added] Iatrogenic causes: post-cardiac surgery, catheter perforation, pacemaker lead.

2.8 Pathophysiology: Rate > Volume (Slide 16) ★

- Haemodynamic consequences **depend on the RATE of accumulation.**
- ★ **A small rapid accumulation can be more dangerous than a large slow one.** Summary slide 30: **~200 mL acutely** vs **> 1 L if slow.**
- Cascade (slide 16 sketch): 1. **↑ Pressure** on the heart 2. The heart cannot stretch out fully between contractions 3. Chambers don't fill properly 4. **↓ Cardiac output** 5. **Hypotension**

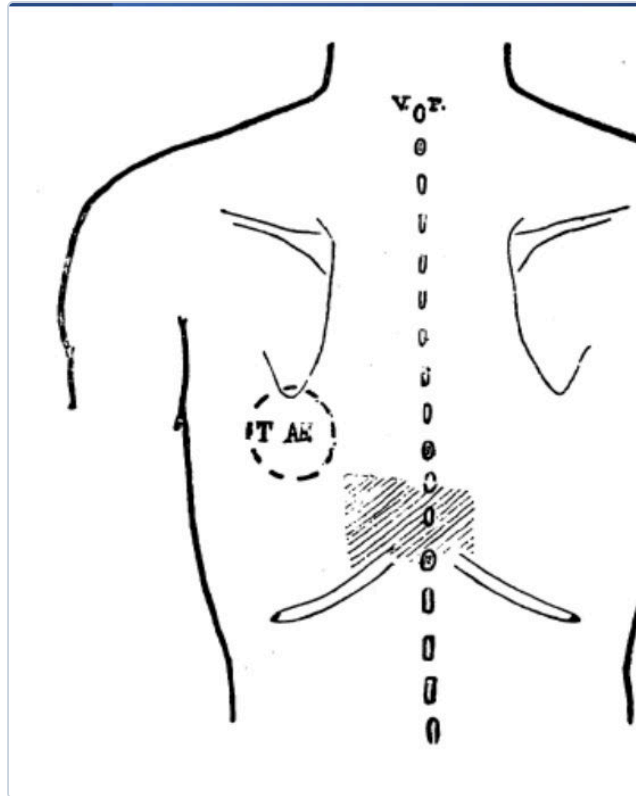


Slide 16: tamponade cascade sketch

2.9 Clinical Features (Slides 17-19)

Effusion (slide 17):

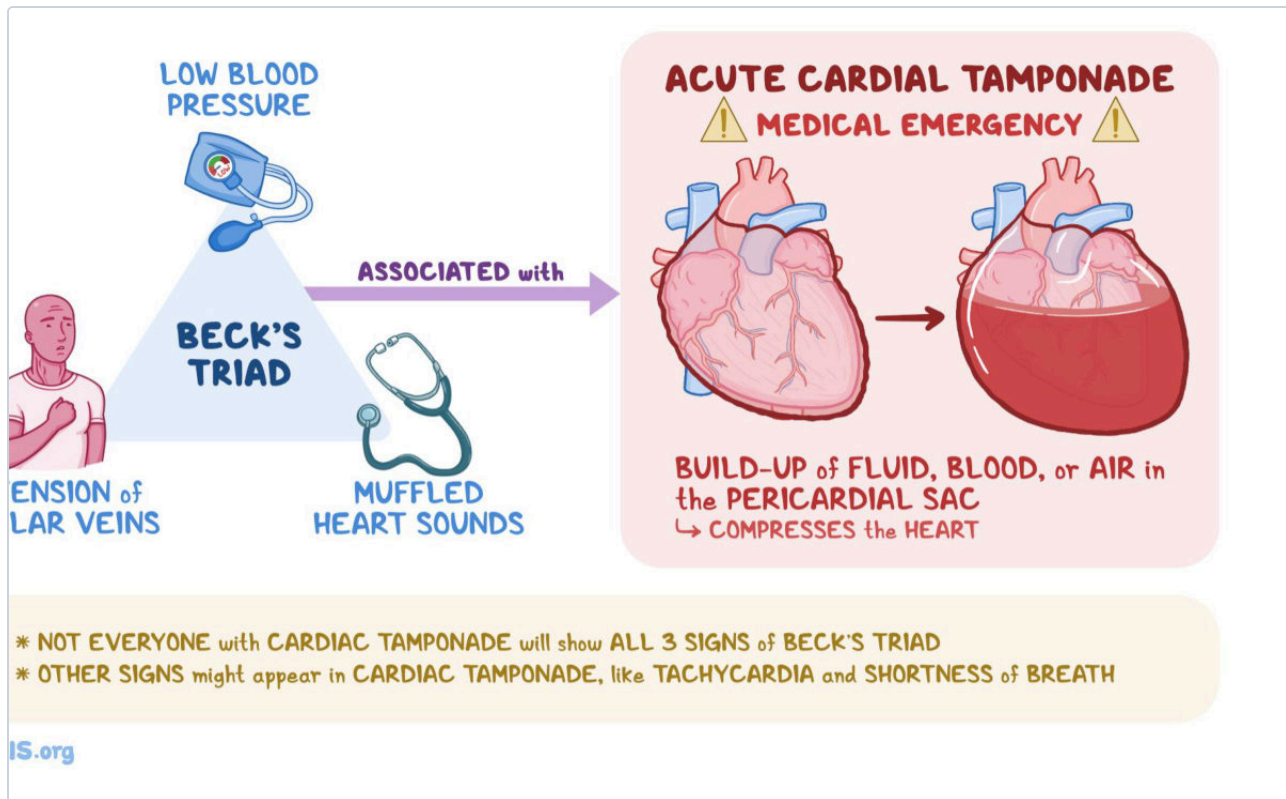
- Dyspnoea, orthopnoea, chest pain
- **Compressive symptoms: hoarseness** (recurrent laryngeal), cough, **hiccups** (phrenic)
- **Apical impulse difficult to locate**
- ★ **Ewart sign**, from compression of lung parenchyma at the **base of the left lung**: - **Dullness to percussion** - **↑ Vocal fremitus** - **Bronchial breathing**



Slide 17: Ewart sign area (left lung base)

Tamponade (slide 18):

- ★ **Beck's triad:** - Hypotension - Muffled heart sounds - Distended neck veins
- **Tachycardia**
- **Tachypnoea**
- ★ **Pulsus paradoxus**
- Slide 18 note: **not everyone with tamponade shows all 3 signs** of Beck's triad. Tachycardia and dyspnoea may be the clues.

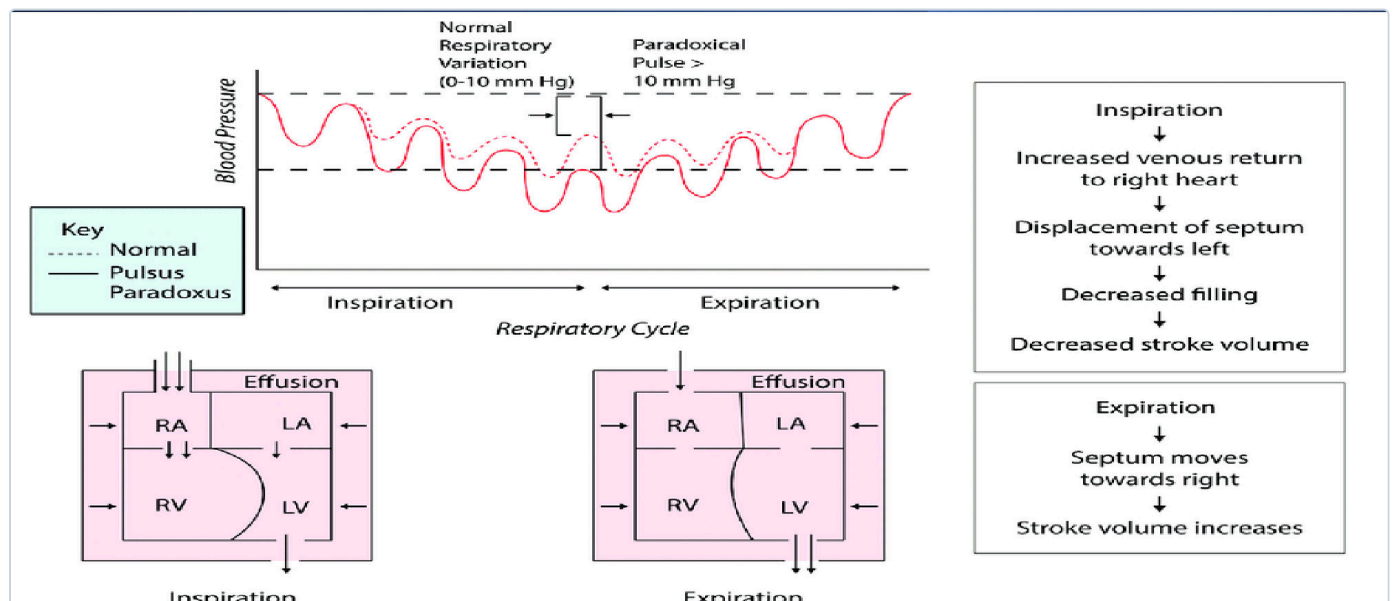


Slide 18: Beck's triad

Mnemonic: Beck's = "3 D's" [added] → Distant heart sounds · Distended neck veins · Decreased BP.

Pulsus paradoxus (slide 19):

- **Definition:** inspiratory fall in SBP > 10 mmHg (normal respiratory variation is 0–10 mmHg).
- **Inspiration:** 1. ↑ Venous return to the right heart 2. **The septum is displaced towards the left** 3. LV filling ↓ 4. **Stroke volume ↓**
- **Expiration:** the septum moves back to the right and stroke volume increases.



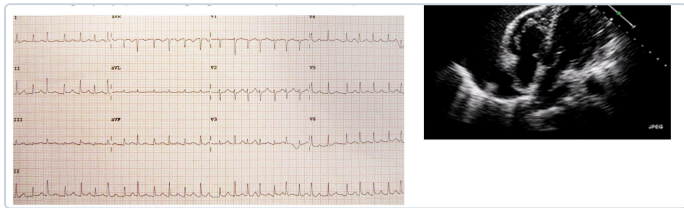
Slide 19: mechanism of pulsus paradoxus

[added] Other causes of pulsus paradoxus: severe asthma/COPD, massive PE, constrictive pericarditis (uncommon), hypovolaemic shock. In tamponade the y-descent is absent on the JVP; in constriction it

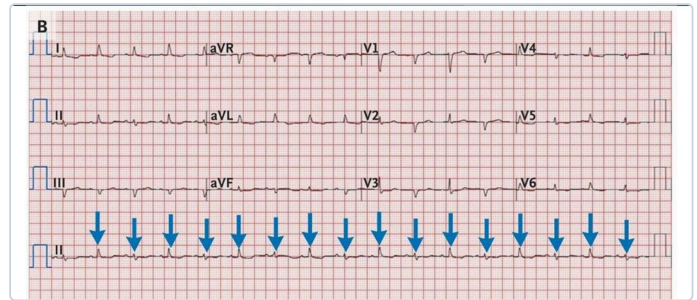
is steep.

2.10 Diagnostics (Slides 20-22) ★

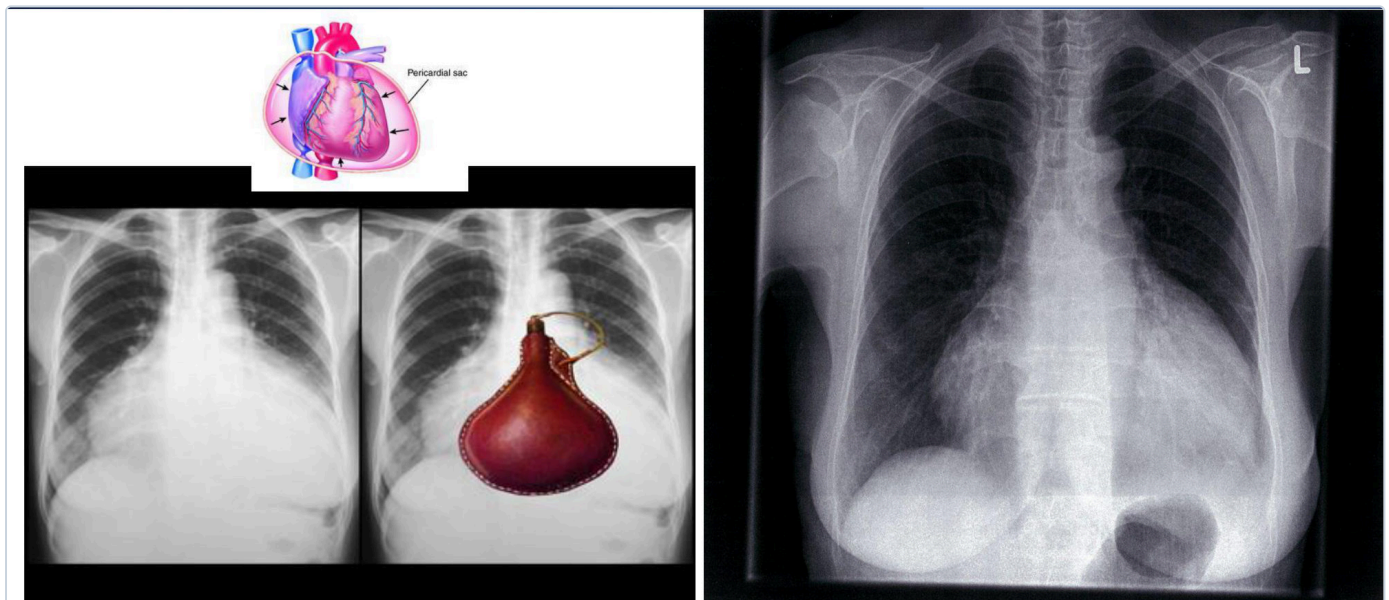
Test	Finding
ECG	Low-voltage QRS + ★ electrical alternans (beat-to-beat QRS amplitude change as the heart swings in the fluid)
CXR	Enlarged cardiac silhouette: ★ "water-bottle" heart (needs > ~200 mL [added])
Echocardiography	★ Imaging modality of choice. [added] Tamponade signs: RA collapse in systole , RV collapse in diastole , plethoric IVC



Slide 20: low voltage ECG and echo of effusion



Slide 21: electrical alternans (arrows)

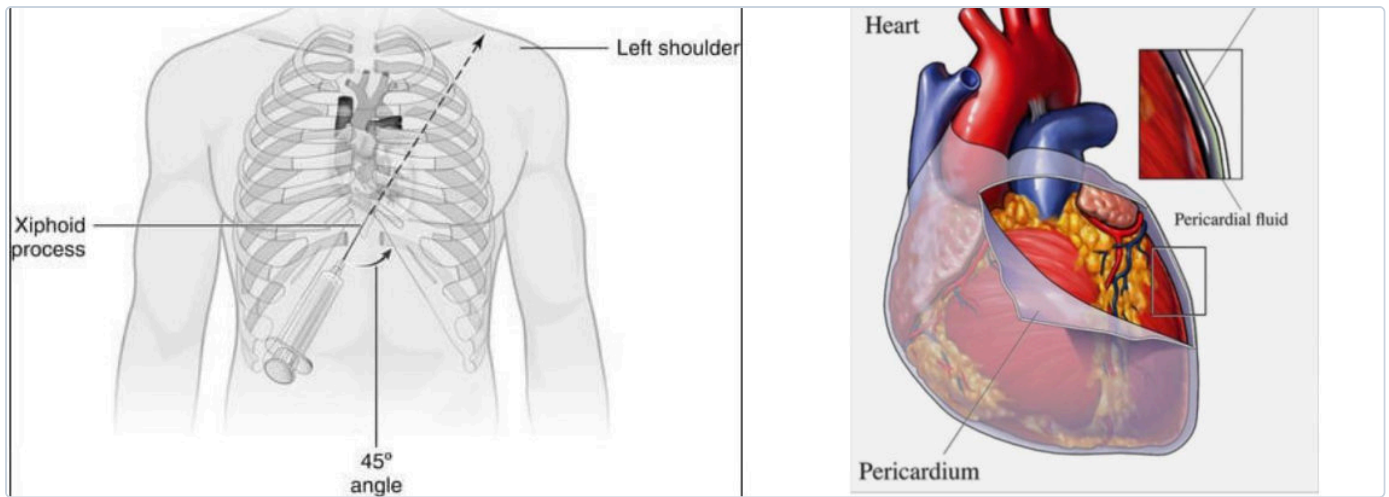


Slide 22: water-bottle heart on CXR

2.11 Management (Slide 23) ★

- **Routine drainage is unnecessary without haemodynamic compromise.** Treat the cause.
- ★ **Tamponade is a life-threatening emergency** → **urgent drainage** (pericardiocentesis).
- ★ **Fluid resuscitation** ↑ **preload** and filling of the cardiac chambers.
- **Chronic recurrent effusions** → surgical **pleuropericardial window**.

Pericardiocentesis (slide 23 figure): subxiphoid approach, needle at a **45° angle aimed at the left shoulder**.



Slide 23: subxiphoid pericardiocentesis

[added] Avoid diuretics, nitrates and positive-pressure ventilation in tamponade, because each ↓ preload and can precipitate arrest. Traumatic or post-MI rupture with clot needs surgery rather than a needle.

🔁 **Recap: Effusion and tamponade.** Fluid in the pericardial sac; rate beats volume. Effusion signs: hard-to-find apex, Ewart sign. Tamponade: Beck's triad (↓ BP, muffled sounds, ↑ JVP), tachycardia, pulsus paradoxus > 10 mmHg. ECG: low voltage + electrical alternans. CXR: water-bottle heart. Echo is the test of choice. Tamponade → urgent pericardiocentesis + IV fluids; asymptomatic → don't drain; recurrent → pericardial window.

🔗 **Link:** If inflammation heals with fibrosis instead of fluid, the pericardium becomes a rigid shell. The filling problem is the same, but it is chronic.

👉 **Up next: Constrictive pericarditis.**

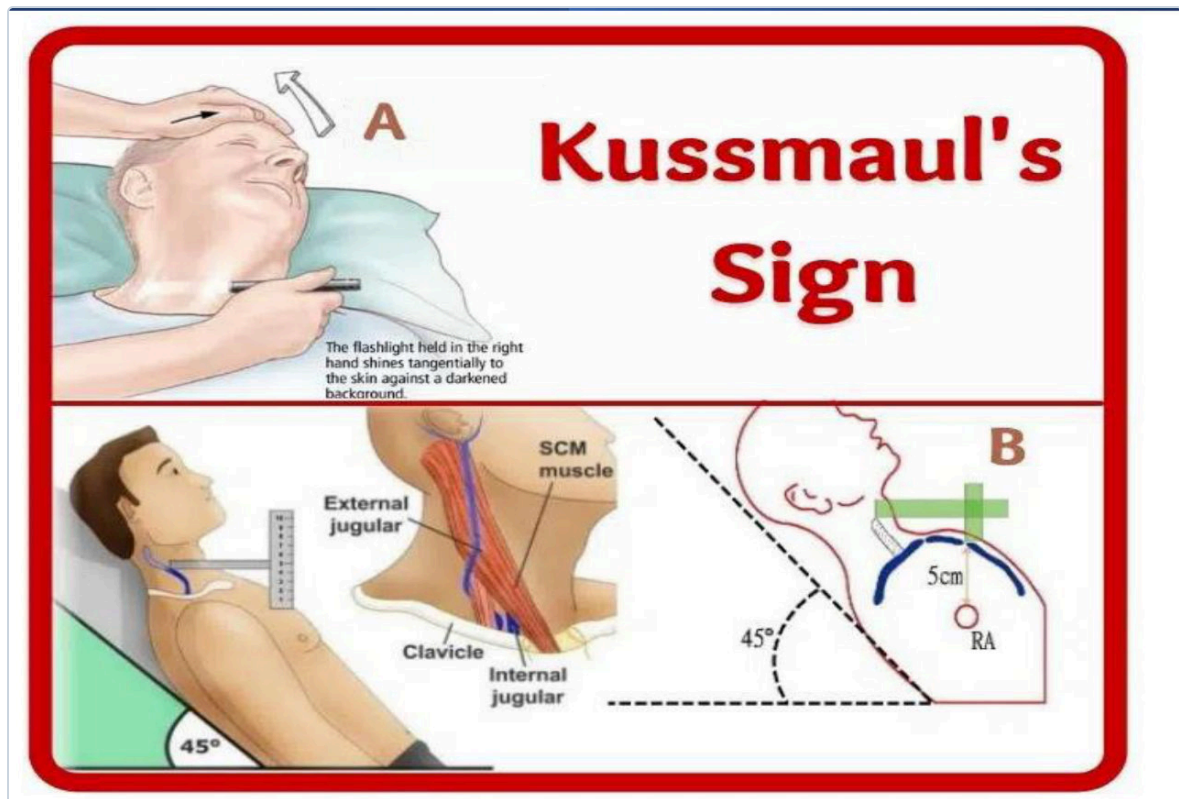
2.12 Constrictive Pericarditis: Definition & Etiology (Slides 24-25)

- **Definition:** pericardial constriction from **inflammation → scarring, fibrosis and adhesion of the pericardial layers**.
- **Mechanism:** a **rigid, scarred pericardium limits diastolic filling** of the ventricles → ↑ **intracardiac pressures**.
- **Etiology:** - Infections - ★ **Prior cardiac surgery** - Trauma - ★ **Irradiation** - Connective tissue disorders - Uraemia - Neoplastic involvement - ★ **Tuberculous pericarditis**

[added] TB is the commonest cause in developing countries and an exam favourite. Post-surgery and radiation are the commonest causes in developed countries.

2.13 Clinical Features (Slide 26) ★

- **Fatigue, dyspnoea on exertion**
- **Lower-extremity oedema, ★ ascites** (often out of proportion to the oedema [added])
- Tachycardia
- **JVD**
- ★ **Kussmaul sign: JVP rises (or fails to fall) on inspiration**
- ★ **Pericardial knock:** early diastolic sound when ventricular filling stops abruptly

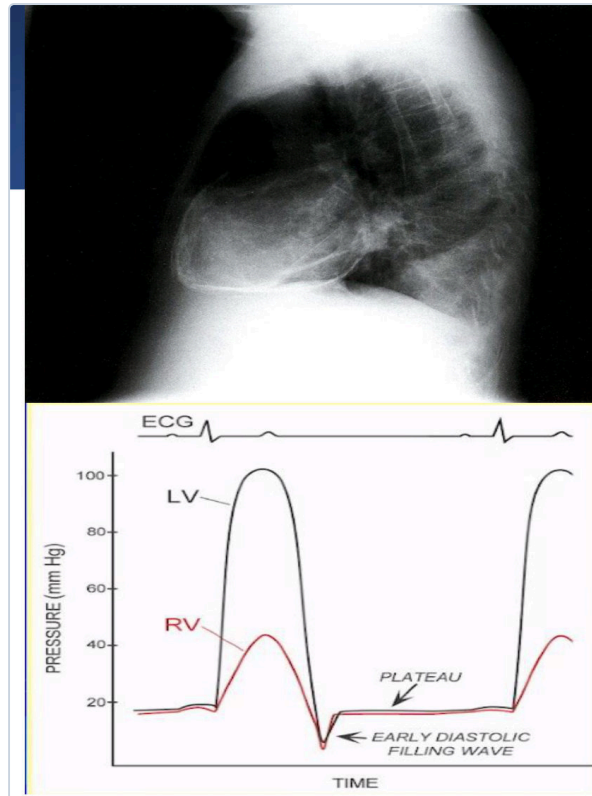


Slide 26: Kussmaul's sign and JVP assessment at 45°

[added] Patients present like right heart failure with clear lungs, and are often mistaken for cirrhosis. Clue: ↑JVP, which cirrhosis lacks. Hepatomegaly is common.

2.14 Diagnostics (Slides 27-28)

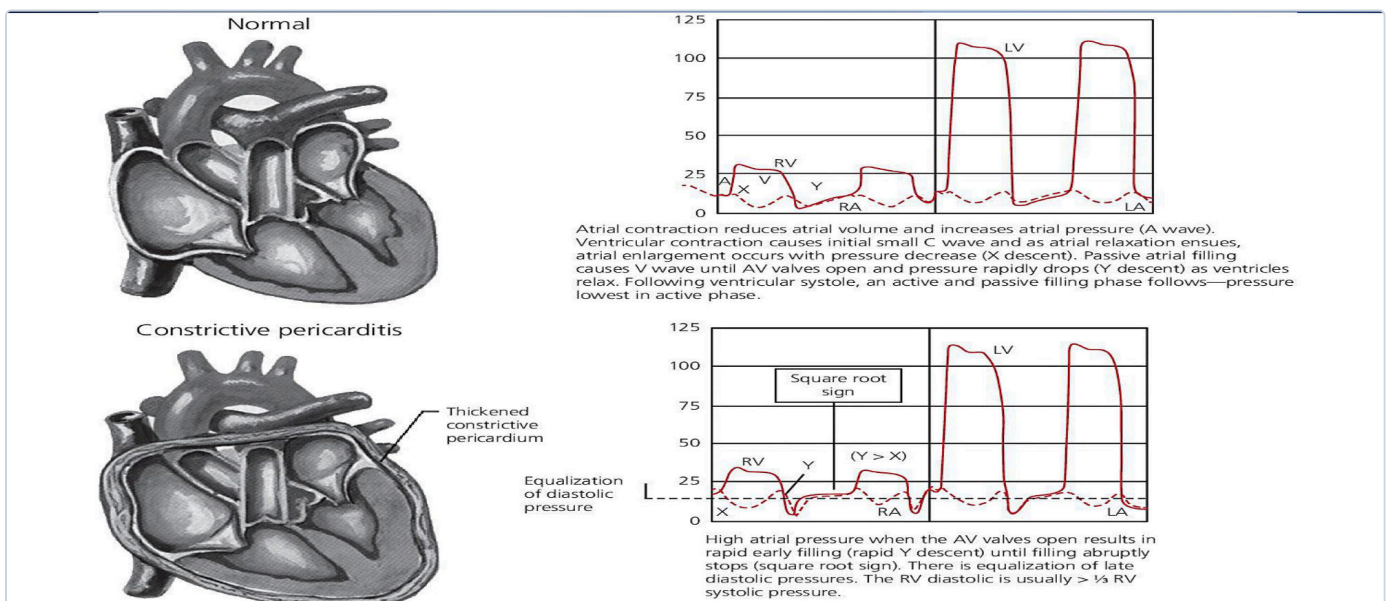
Test	Finding
ECG	Low QRS voltage, left atrial enlargement ([added] ± AF)
CXR	★ Pericardial calcification (best seen on the lateral film)
Echo	↑ Pericardial thickness
Cardiac CT/MRI	Thickened/calcified pericardium ([added] > 4 mm)
Cardiac catheterisation	↑ ventricular diastolic pressures · ★ equalisation of diastolic pressures across chambers · ★ "dip and plateau" ("square-root sign")



Slide 27: pericardial calcification on lateral CXR and dip-and-plateau trace

Slide 28 explained:

- **Normal heart:** atrial contraction gives the **A wave**. The initial small C wave and atrial relaxation give the **X descent**. Passive atrial filling gives the **V wave**. AV valve opening gives the **Y descent**.
- **Constriction:** 1. **High atrial pressure when the AV valves open → rapid early filling (rapid Y descent, $Y > X$)**. 2. Filling then **stops abruptly** at the stiff shell (**square-root sign**). 3. **Late diastolic pressures equalise**. 4. **RV diastolic pressure is usually $> \frac{1}{3}$ of RV systolic pressure**.



Slide 28: normal vs constrictive haemodynamics (square-root sign)

2.15 Management (Slide 29)

- ★ **Pericardiectomy is the ONLY definitive treatment.**
- **Na restriction + diuretics** for non-surgical candidates.

[added] For TB pericarditis: anti-TB therapy ($\pm$ steroids) to reduce progression to constriction.

Recap: Constriction. A rigid, scarred, often calcified pericardium limits diastolic filling. Causes: TB, surgery, radiation, CTD, uraemia, cancer. Right-heart-failure picture with ascites, $\uparrow$ JVP, Kussmaul sign, pericardial knock. ECG low voltage + LAE; CXR calcification; echo/CT/MRI thick pericardium. Cath: equal $\uparrow$ diastolic pressures with dip-and-plateau. Pericardiectomy is the only cure; diuretics + salt restriction if the patient is not fit for surgery.

2.16 Lecturer Summary (Slides 30-31)

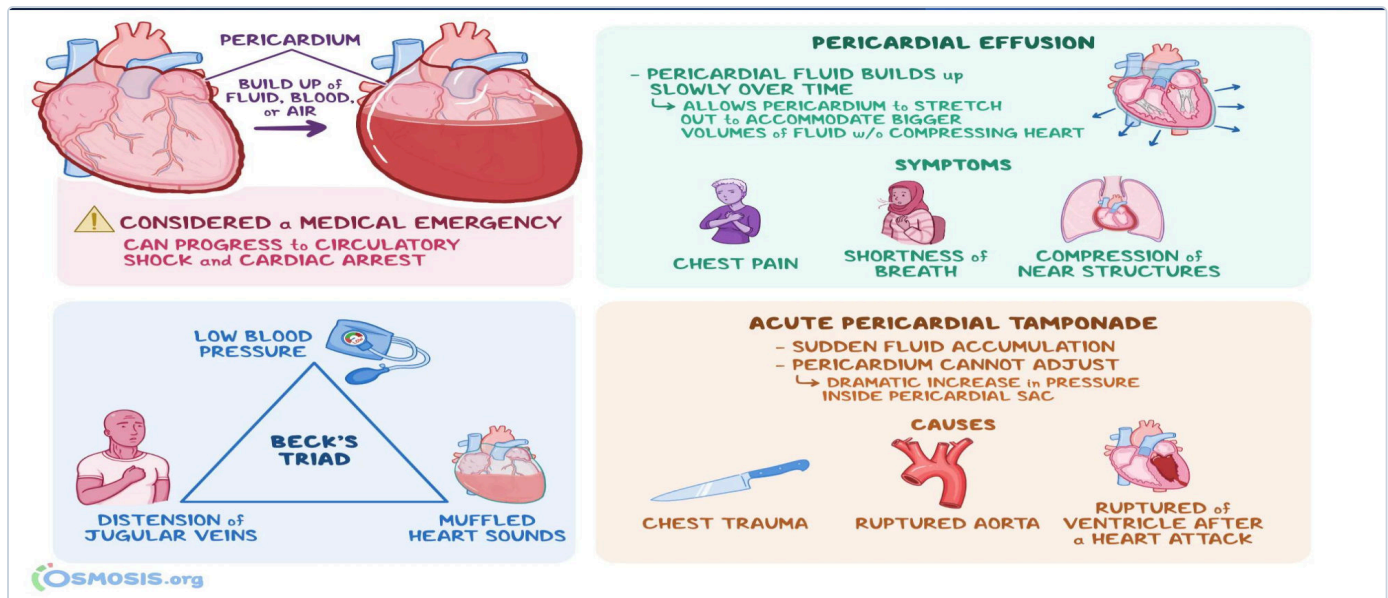
	Diagnosis	Treatment
Acute pericarditis (inflammation of the pericardium)	$\uparrow$ WBC, ESR, CRP; ECG: ST elevation in several leads, PR depression	Anti-inflammatory drugs , can combine with colchicine
Pericardial effusion (200 mL acutely, > 1 L if slowly)	Small $\rightarrow$ asymptomatic, unless associated with pericarditis. Large $\rightarrow$ cardiac tamponade	Small $\rightarrow$ no treatment. Tamponade $\rightarrow$ hospital admission, pericardiocentesis, volume expansion
Constrictive pericarditis (fibrinous scarring and calcification)		Pericardiectomy

ACUTE PERICARDITIS	DIAGNOSIS	TREATMENT
INFLAMMATION of the PERICARDIUM 	$\uparrow$ WBC, ESR, & CRP - ECG - ST SEGMENT ELEVATION in SEVERAL LEADS - PR SEGMENT DEPRESSION	- ANTI-INFLAMMATORY MEDICATIONS - can COMBINE w/ COLCHICINE
PERICARDIAL EFFUSION		
200 mL of FLUID ACUTELY > 1 L if SLOWLY 	SMALL - ASYMPTOMATIC - UNLESS ASSOCIATED w/ PERICARDITIS LARGE - CARDIAC TAMPONADE	SMALL $\rightarrow$ NO TREATMENT HOSPITAL ADMISSION PERICARDIOCENTESIS VOLUME EXPANSION
CONSTRUCTIVE PERICARDITIS		
FIBRINOUS SCARRING & CALCIFICATION of the PERICARDIUM 		PERICARDECTOMY

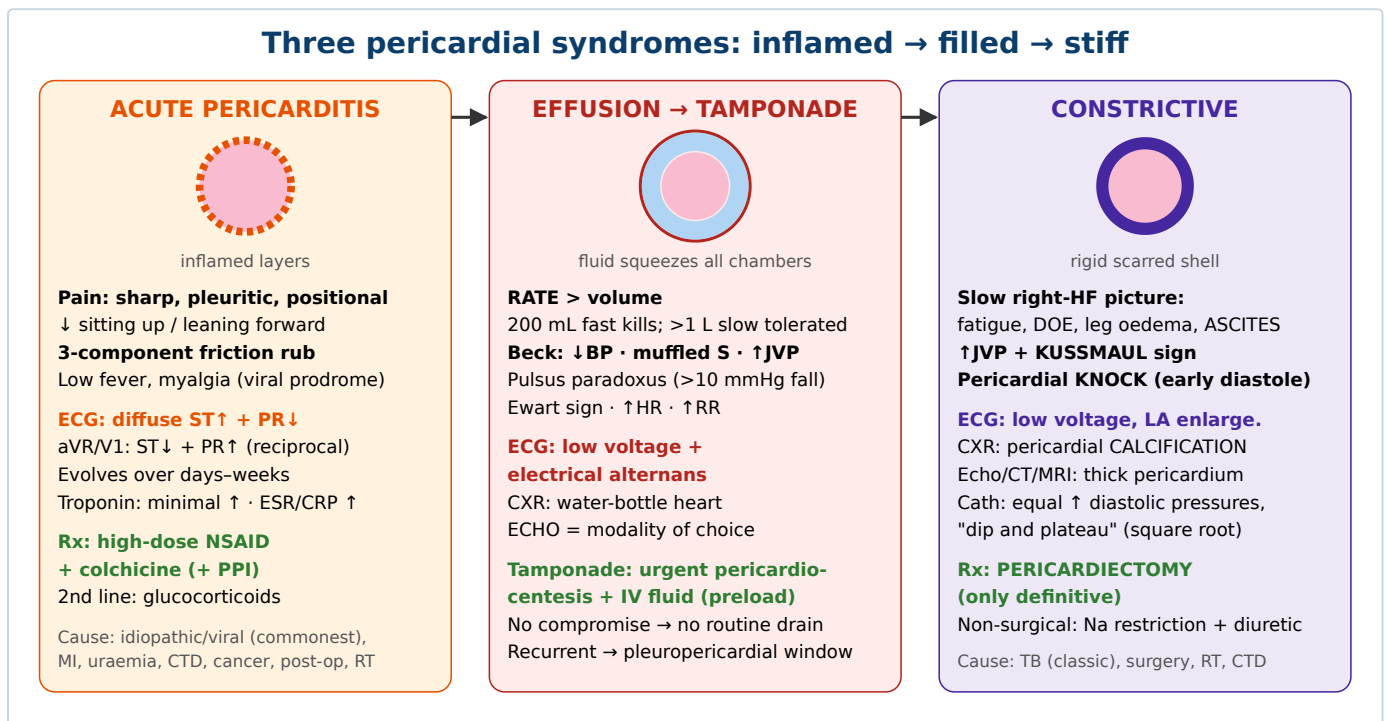
Slide 30: lecturer summary table

Slide 31 (Osmosis):

- A **pericardial effusion** builds slowly, so the pericardium stretches and symptoms are chest pain, shortness of breath and compression of nearby structures.
- Acute tamponade** is **sudden** fluid accumulation into a sac that cannot adapt. - Causes: **chest trauma, ruptured aorta, ventricular rupture after heart attack**. - It **can progress to circulatory shock and cardiac arrest**.



Slide 31: effusion vs acute tamponade



Added diagram: the three pericardial syndromes side by side

3. High-Yield Comparison Tables

Table A: Pericarditis vs STEMI ★

Feature	Acute pericarditis	STEMI
Pain	Sharp, pleuritic, positional (better leaning forward)	Pressure/heaviness, not positional
Exam	Friction rub	± S4, new MR murmur
ST elevation	Diffuse, concave	Territorial, convex
Reciprocal ST↓	Only aVR (± V1)	Yes , in opposite leads

Feature	Acute pericarditis	STEMI
PR segment	Depressed (↑ in aVR)	Normal
Evolution	Days-weeks ; T inverts after ST normalises	Hours ; Q waves; T inverts while ST still ↑
Troponin	Minimal ↑	Marked rise and fall
Echo	± effusion; normal wall motion	Regional wall-motion abnormality
Treatment	NSAID + colchicine	Reperfusion (PCI/lysis), antiplatelets

Table B: Tamponade vs Constrictive Pericarditis ★

Feature	Cardiac tamponade	Constrictive pericarditis
Onset	Acute	Chronic
Pericardium	Fluid-filled	Thick, fibrotic, calcified
Haemodynamics	Compression throughout the cardiac cycle	Early filling is free, then stops abruptly
BP	Hypotension (Beck)	Usually normal/low
JVP	↑	↑
JVP y-descent [added]	Absent/blunted	Steep (prominent)
★ Signature sign	Pulsus paradoxus	★ Kussmaul sign
Heart sounds	Muffled	Pericardial knock
Ascites/oedema	Not prominent	Prominent ascites , oedema
ECG	Low voltage + electrical alternans	Low voltage + LA enlargement
CXR	Water-bottle heart	Pericardial calcification
Echo	Effusion; RA/RV diastolic collapse [added]	Thickened pericardium ; septal bounce [added]
Cath	Equalised diastolic pressures [added]	Equalised diastolic pressures + dip and plateau
Treatment	Urgent pericardiocentesis + fluids	Pericardiectomy (diuretics if not a surgical candidate)

Table C: Signs Dictionary

Sign	Definition	Disease
Beck's triad	Hypotension + muffled heart sounds + distended neck veins	Tamponade
Pulsus paradoxus	SBP falls > 10 mmHg on inspiration	Tamponade (also severe asthma, PE)
Ewart sign	Dullness + ↑ fremitus + bronchial breathing at the left lung base	Large effusion compressing the lung
Kussmaul sign	JVP rises on inspiration	Constriction ([added] also RV infarct, restrictive CMP, massive PE)
Pericardial knock	Early diastolic sound	Constriction
Friction rub (3 components)	Ventricular systole + ventricular diastole + atrial systole	Acute pericarditis
Electrical alternans	Alternating QRS amplitude	Large effusion/tamponade
Water-bottle heart	Globular, enlarged silhouette	Large effusion

Sign	Definition	Disease
Square-root sign	Dip-and-plateau ventricular diastolic pressure	Constriction

Table D: Treatment Ladder

Condition	First	If fails / special
Acute pericarditis	High-dose NSAID/aspirin + colchicine + PPI	Glucocorticoids (2nd line); aspirin post-MI <i>[added]</i>
Recurrent pericarditis	Colchicine (CORP) + NSAID	Steroids; <i>[added]</i> anakinra/rilonacept
Effusion, no compromise	No routine drainage ; treat the cause	
Tamponade	★ Urgent pericardiocentesis + IV fluids	Surgery if rupture/clot
Recurrent effusions		Pleuropericardial window
Constriction	★ Pericardiectomy	Na restriction + diuretics if not a surgical candidate

4. Rules, Exceptions, Patterns

Rule	Exception / nuance
Pericarditis has ST elevation in many leads	aVR (± V1) shows ST depression + PR elevation (the reciprocal mirror)
Troponin ↑ suggests MI	In pericarditis troponin is minimally ↑ (epicardial inflammation). Marked ↑ = myopericarditis <i>[added]</i>
NSAIDs for pericarditis	Post-MI → aspirin (NSAIDs impair healing) <i>[added]</i>
Steroids are anti-inflammatory	They are second line because they increase recurrence <i>[added]</i>
Bigger effusion = sicker	★ Rate beats volume : 200 mL fast can kill; > 1 L slow may be tolerated
Hypotension + ↑JVP = give diuretics?	★ NO in tamponade. Give fluids (↑ preload)
Drain effusions	Not routinely if haemodynamically stable
Beck's triad defines tamponade	Not all 3 are always present (slide 18)
Pulsus paradoxus = tamponade	Also severe asthma/COPD, massive PE <i>[added]</i>
Kussmaul = constriction	Also RV infarct, restrictive CMP <i>[added]</i> ; rare in tamponade
Constriction is medical	Pericardiectomy is the only definitive treatment

5. Clinical Correlations

- **Young man with a viral prodrome, pleuritic pain better leaning forward, rub, diffuse ST↑/PR↓** → viral pericarditis → NSAID + colchicine (opening case).
- **Day 2-4 after a transmural MI with pleuritic pain** → **early post-MI pericarditis**. **Weeks after MI or cardiac surgery** → **Dressler / post-cardiac injury syndrome** (autoimmune; 33% in the slide 6 cohort). *[CAD lecture link]*
- **Patient 5 days after MI with sudden PEA, ↑JVP and muffled sounds** → **free wall rupture** → **tamponade**.

- **Stab wound to the chest + hypotension + ↑JVP + muffled sounds** → tamponade → pericardiocentesis/surgery.
- **Uraemic patient with a rub** → **uraemic pericarditis** → [added] urgent **dialysis** is an indication.
- **Hypothyroid patient with an enlarged cardiac silhouette** → effusion (usually slow, rarely tamponade).
- **Lung or breast cancer patient with dyspnoea, low-voltage ECG and alternans** → malignant effusion/tamponade → drain, then window if recurrent.
- **Immigrant with prior TB, ascites, ↑JVP that rises on inspiration, and pericardial calcification** → **TB constrictive pericarditis** → pericardiectomy.
- **Post-mediastinal radiation (lymphoma) years later** → constriction.
- [added] **Hypotension + ↑JVP + clear lungs** differential: **tamponade, tension pneumothorax, massive PE, RV infarct**. Echo separates them.

6. Rapid Revision (Last-Day)

1. Pericardium: **fibrous** outer bag (attached to the diaphragm) + **serous** parietal/visceral layers; **< 50 mL** fluid normally.
2. Acute pericarditis: **idiopathic/viral** commonest; also MI, uraemia, CTD, cancer, post-op, radiation, trauma.
3. **Positional pleuritic pain**, better leaning forward; **3-component rub**; fever, hiccups.
4. ECG: **diffuse ST↑ + PR↓, reciprocal ST↓ + PR↑ in aVR/V1**, evolving over **days-weeks**.
5. vs STEMI: **diffuse / days-weeks / PR depressed** vs **localised / hours / PR normal**.
6. Labs: WBC, ESR, CRP, CK ↑; **troponin minimal**.
7. Rx: **high-dose NSAID + colchicine + gastroprotection**; steroids 2nd line. **ICAP** (first episode) and **CORP** (recurrent) prove colchicine works.
8. Tamponade = ↑ intrapericardial pressure → ↓ filling → ↓ CO. **Rate > volume** (200 mL fast vs > 1 L slow).
9. Effusion: hard-to-find apex, **Ewart sign**, compressive hoarseness and hiccups.
10. **Beck's triad**: hypotension, muffled sounds, distended neck veins (+ ↑HR, ↑RR, **pulsus paradoxus > 10 mmHg**).
11. ECG **low voltage + electrical alternans**; CXR **water-bottle**; **echo is the test of choice**.
12. No compromise → no routine drainage. **Tamponade → urgent pericardiocentesis (subxiphoid, 45°, towards the left shoulder) + IV fluids**. Recurrent → **pleuropericardial window**.
13. Constriction: scarred, rigid pericardium; **TB, surgery, radiation**.
14. **Kussmaul sign, pericardial knock, ascites**, ↑JVP, leg oedema.
15. ECG low voltage + **LAE**; CXR **calcification**; echo/CT/MRI **thickening**; cath **equal diastolic pressures + dip-and-plateau (square root)**.
16. **Pericardiectomy is the only definitive treatment**; salt restriction + diuretics if not operable.

7. Exam-Style Questions

Q1 (lecture opening case). A 23-year-old man has 1 week of sharp substernal pain, worse with inspiration and relieved by leaning forward, plus nausea and myalgia. He has a high-pitched rub at the LSB and an ECG with PR depression and diffuse ST elevation. What is the most likely cause? a. Dressler syndrome · b. Acute MI · c. *M. tuberculosis* · d. SLE · e. Acute viral infection

Answer: e. Acute viral infection. *[reasoning added; key not on slides]*

- Young patient with a **viral prodrome (myalgia, nausea)** and classic pericarditis → **idiopathic/viral** is the commonest cause.
- Wrong options:
 - **Dressler** needs a preceding MI or cardiac surgery.
 - **Acute MI** gives localised convex ST↑ without PR depression; his father's CAD is a distractor.
 - **TB** is subacute with constitutional symptoms and an endemic exposure.
 - **SLE** would need other systemic features.

Q2. A 60-year-old man has had dyspnoea for 2 days. BP 82/60, HR 122, JVP raised, heart sounds faint, lungs clear. SBP falls by 18 mmHg on inspiration. The ECG shows low-voltage QRS with alternating amplitude. What is the most appropriate immediate management? A. IV furosemide · B. IV nitroglycerin · C. Urgent pericardiocentesis with IV fluid · D. NSAID + colchicine · E. Pericardiectomy

Answer: C.

- **Beck's triad + pulsus paradoxus + electrical alternans = cardiac tamponade**, a life-threatening emergency.
- Management is **urgent drainage**, and **fluid resuscitation supports preload** (slide 23).
- **Diuretics and nitrates** drop preload and can cause arrest *[added]*.
- **Pericardiectomy** is for constriction.

Q3. A 45-year-old man who had TB pericarditis 6 years ago now has ascites, leg oedema and fatigue. His JVP rises on inspiration, and an early diastolic sound is heard. The lateral CXR shows a rim of calcification around the heart. What is the definitive treatment? A. Pericardiocentesis · B. Pericardial window · C. Pericardiectomy · D. Colchicine · E. Diuretics + salt restriction

Answer: C.

- **Kussmaul sign + pericardial knock + calcification** after TB = **constrictive pericarditis**.
- **Pericardiectomy is the only definitive treatment** (slide 29).
- Diuretics and salt restriction (E) are only for **non-surgical candidates**.
- There is no fluid to drain, so A and B are wrong.

Q4. Which ECG feature best distinguishes acute pericarditis from acute anterior STEMI? A. ST elevation in V2–V4 · B. **PR-segment depression with diffuse ST elevation evolving over days** · C. Hyperacute T waves · D. Pathological Q waves · E. Raised troponin

Answer: B (slide 12).

- Pericarditis is **diffuse** with **PR depression**, evolving over **days-weeks**.
- STEMI is **localised** with a **normal PR**, evolving over **hours**, and produces Q waves.
- Troponin can rise in both (minimal in pericarditis).

Q5. Which statement about pericardial effusion is correct? A. Every effusion > 500 mL should be drained · B. Tamponade depends only on effusion volume · C. **A small, rapidly accumulating effusion can cause tamponade while a large, slow one may not** · D. CXR is the imaging modality of choice · E. Recurrent effusions are best treated by repeated pericardiocentesis

Answer: C (slide 16).

- A is wrong: **no routine drainage** without haemodynamic compromise.
- D is wrong: **echo** is the modality of choice.
- E is wrong: recurrent effusions need a **pleuropericardial window**.

8. Exam Pitfalls & Traps

- **Leaning forward relieves pericarditis pain.** Lying supine worsens it. Don't swap these.
- **Diffuse ST elevation $\neq$ STEMI.** Look for **PR depression** and the **aVR mirror**. Thrombolysing pericarditis risks **haemopericardium** *[added]*.
- **Troponin positive does not exclude pericarditis.** It is minimally raised (slide 8).
- **Dressler vs viral:** Dressler requires a **preceding MI or cardiac surgery** (weeks before). Without that history, choose viral.
- **"3 components" of the rub** = ventricular systole, ventricular diastole, **atrial systole**. Atrial systole is lost in **AF**, leaving 2 components *[added]*.
- **Tamponade + hypotension $\rightarrow$ fluids, NOT diuretics**, even with a high JVP.
- **Beck's triad can be incomplete** (slide 18). Don't exclude tamponade because one sign is missing.
- **Pulsus paradoxus = tamponade; Kussmaul = constriction.** This is the classic swap trap. Kussmaul can appear in effusive-constrictive disease but is **not** a tamponade sign *[added]*.
- **Ewart sign** is a **lung** finding (left base: dullness, bronchial breathing). It is easily mistaken for left lower lobe pneumonia.
- **Electrical alternans** = tamponade or a large effusion. **Low voltage alone** is nonspecific: effusion, constriction, obesity, COPD, hypothyroidism *[added]*.
- **Calcification** on CXR $\rightarrow$ **constriction**, not tamponade. **Water-bottle heart** $\rightarrow$ **effusion**.
- **Slide 30 numbers:** 200 mL acutely can cause tamponade; **> 1 L** if slow. Normal pericardial fluid is **< 50 mL** (slide 5).
- **Slide 6 etiology chart** shows **post-cardiac injury at 33%**. That is a cohort-specific figure. For "most common cause", the answer is **idiopathic/viral**.
- **Constriction mimics cirrhosis** (ascites, oedema). **$\uparrow$ JVP is the discriminator.**